

SYNOPSIS: CRYPTO ANALYSIS (Mini-data visualisation project)

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Introduction:

The Excel sheet contains historical data of Bitcoin, one of the most popular cryptocurrencies in the world. The dataset includes important variables such as Date, Open price, High price, Low price, Closing price, Trading Volume, and Daily Percentage Change. This data represents the daily performance of Bitcoin over a period of time and is widely used for financial and market analysis.

Analyzing this Excel dataset helps in understanding price trends, market volatility, and trading behavior of Bitcoin. By organizing and cleaning the data, meaningful insights can be obtained through graphs and statistical analysis. Such datasets are commonly used in data analytics, financial forecasting, investment analysis, and academic projects. The Excel sheet serves as a foundation for performing data visualization, trend analysis, and decision-making using tools like Excel and R programming.

The real dataset contains data of 2010-2020 but for the sake of simplicity the dataset used here consists of recent 4 years of data(2017-2020)

Real Life Implementations

1 Where can this Bitcoin data be used in real life?

This dataset can be used wherever **price behaviour, trends, and risk** need to be analyzed.

1. Investment analysis

- Investors analyze price trends to decide buy / sell / hold.
- Helps identify bullish and bearish periods.

2. Financial forecasting

- Used to predict future prices using time-series models.
- Helpful for traders and crypto analysts.

3. Risk management

- Daily percentage change shows volatility and risk level.
- Useful for portfolio diversification.

4. Trading strategy design

- High, Low, Open, Close prices are used in technical analysis.
- Indicators like moving average, RSI, etc. are built from this data.

5. Market behavior analysis

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- Volume data helps understand market participation.
- Detects panic selling or heavy buying.

6. Academic projects

- Used in college practicals, mini projects, and research papers.
- Common in data visualization and time-series analysis.

7. Business decision-making

- Crypto exchanges analyze this data to optimize trading platforms.

8. Fraud and anomaly detection

- Sudden spikes or crashes can indicate manipulation or abnormal activity.

Insights

1. Bitcoin prices fluctuate sharply, confirming high volatility.
2. The long-term trend shows overall growth despite short-term losses.
3. High-Low gap proves strong intraday price variation.

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4. Volume spikes during major price changes indicate heavy trading activity.
5. Sudden price drops are often followed by partial recovery.
6. Open and Close prices differ significantly during volatile periods.
7. Frequent negative days show market risk and uncertainty.
8. Increased volume during falling prices suggests panic selling.
9. Market behavior is non-linear and speculative.
10. Bitcoin is a high-risk, high-return digital asset.



Decoding the code

1)To load data:

Import dataset->From Text(Readr)->choose file from your pc->import

Function: read.csv loads the data into variable "data"

```
data <- read.csv("Bitcoin Historical Data.csv", stringsAsFactors = FALSE)
```

2)To view structure of dataset:-->optional

```
str(data)
```

3)To view if the dataset is loaded correctly or not(head function prints first 6 rows,we can see if we need any symbol to numeric conversion or not)-->optional

```
head(data)
```

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4) `data$Date` → selects Date column

`as.Date()` → converts text into date format

`"%b %d, %Y"` → tells R the date pattern(Dec 21 2025)

`data$Date <- as.Date(data$Date, format = "%b %d, %Y")`

{`gsub`=Global substitution

syntax:`gsub(pattern, replacement, text)`

pattern → what you want to find

replacement → what you want to replace it with

text → the string or column to work on

}

5)`gsub(" ", "", data$Date)` → removes commas & return char("45, 123" → "45123")

`as.numeric()` → converts character to number

`data$Price <- as.numeric(gsub(",", "", data$Price))`

`data$Open <- as.numeric(gsub(",", "", data$Open))`

`data$High <- as.numeric(gsub(",", "", data$High))`

`data$Low <- as.numeric(gsub(",", "", data$Low))`

6) `"850K" → "850000"`

Converts shorthand volume into full numeric value

`data$Vol. <- gsub("K", "000", data$Vol.)`

`data$Vol. <- gsub("M", "000000", data$Vol.)`

`data$Vol. <- as.numeric(gsub(",", "", data$Vol.))`

`data$Change <- as.numeric(gsub("%", "", data$`Change %`))`

7) Line plot: Date vs Price

```
plot(data$Date, data$Price,
     type = "l",
     col = "blue",
     lwd = 2,
     xlab = "Date",
     ylab = "Price (USD)",
     main = "Bitcoin Closing Price Over Time")
```

8) Line plot: Open and close price vs data

```
plot(data$Date, data$Open,
     type = "l",
     col = "green",
     lwd = 2,
     xlab = "Date",
     ylab = "Price",
     main = "Open vs Close Price")
```

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```
lines(data$Date, data$Price, col = "red", lwd = 2)
```

```
legend("topleft",  
      legend = c("Open", "Close"),  
      col = c("green", "red"),  
      lty = 1)
```

9) Line plot: High and low prices vs date

```
plot(data$Date, data$High,  
      type = "l",  
      col = "purple",  
      lwd = 2,  
      xlab = "Date",  
      ylab = "Price",  
      main = "High and Low Prices")
```

```
lines(data$Date, data$Low, col = "orange", lwd = 2)
```

```
legend("topleft",  
      legend = c("High", "Low"),  
      col = c("purple", "orange"),  
      lty = 1)
```

10) Plotting data vs volume (total amount of Bitcoin traded (bought and sold) during a particular day)

```
plot(data$Date, data$Vol.,  
      type = "l",  
      col = "brown",  
      lwd = 2,  
      xlab = "Date",  
      ylab = "Volume",  
      main = "Bitcoin Trading Volume")
```

12) Plotting date and change(%)

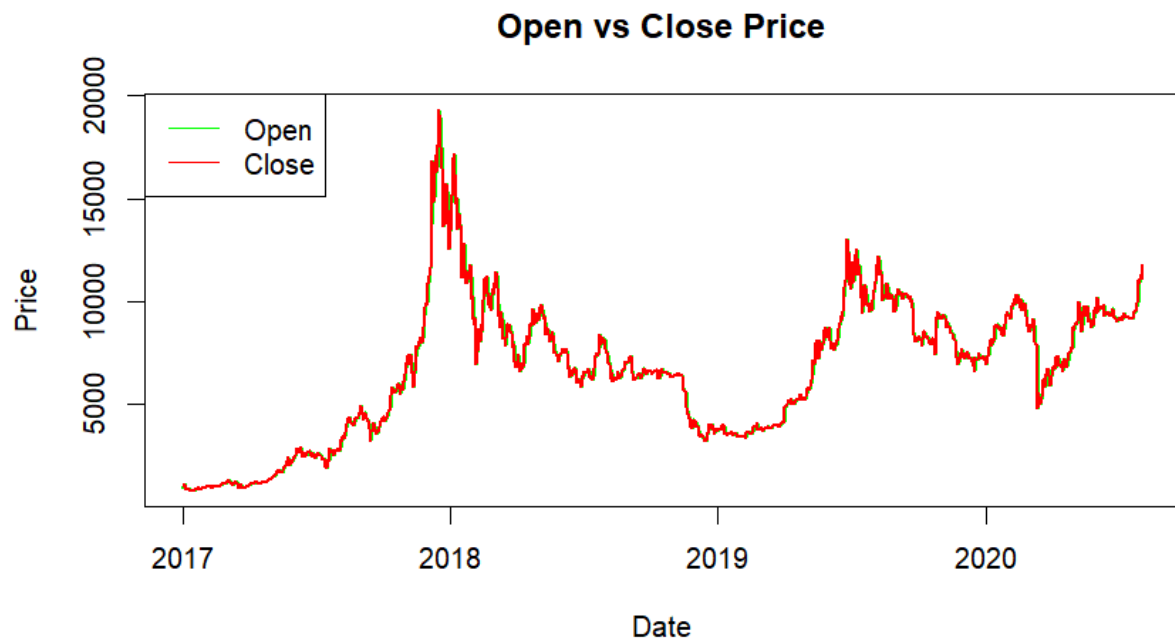
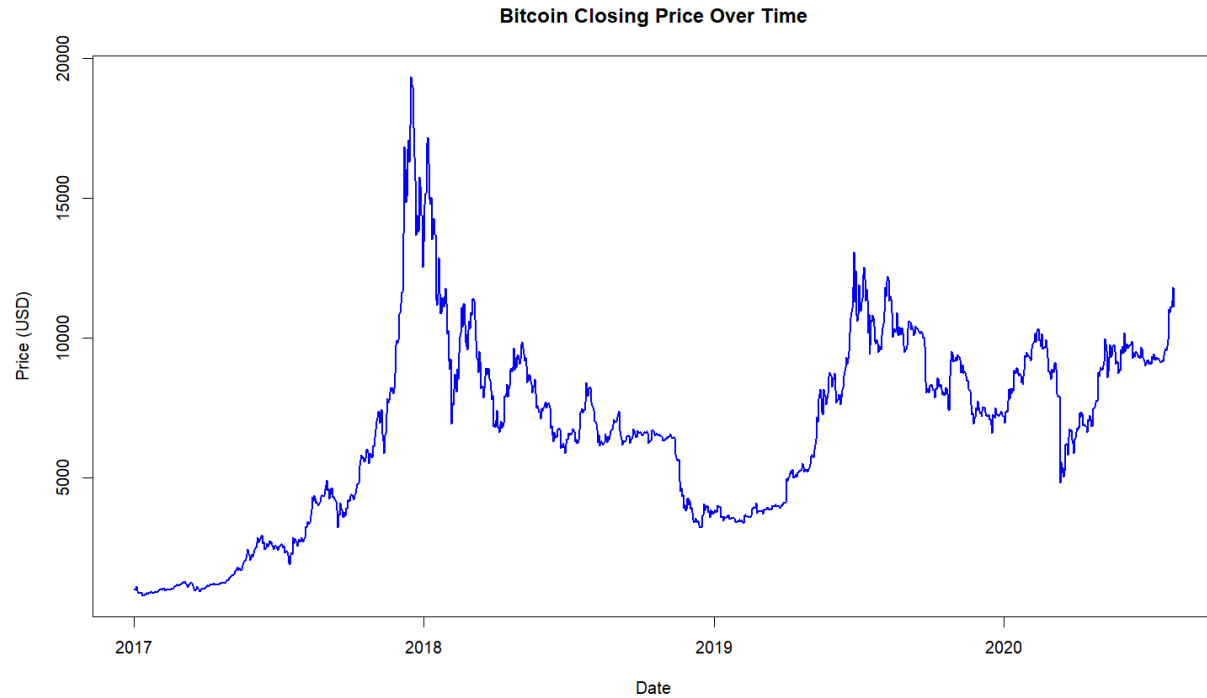
Change refers to the daily percentage change in Bitcoin's price compared to the previous day

```
plot(data$Date, data$Change,  
      type = "h",  
      col = "darkcyan",  
      xlab = "Date",  
      ylab = "Daily Change (%)",  
      main = "Daily Percentage Change in Bitcoin Price")
```

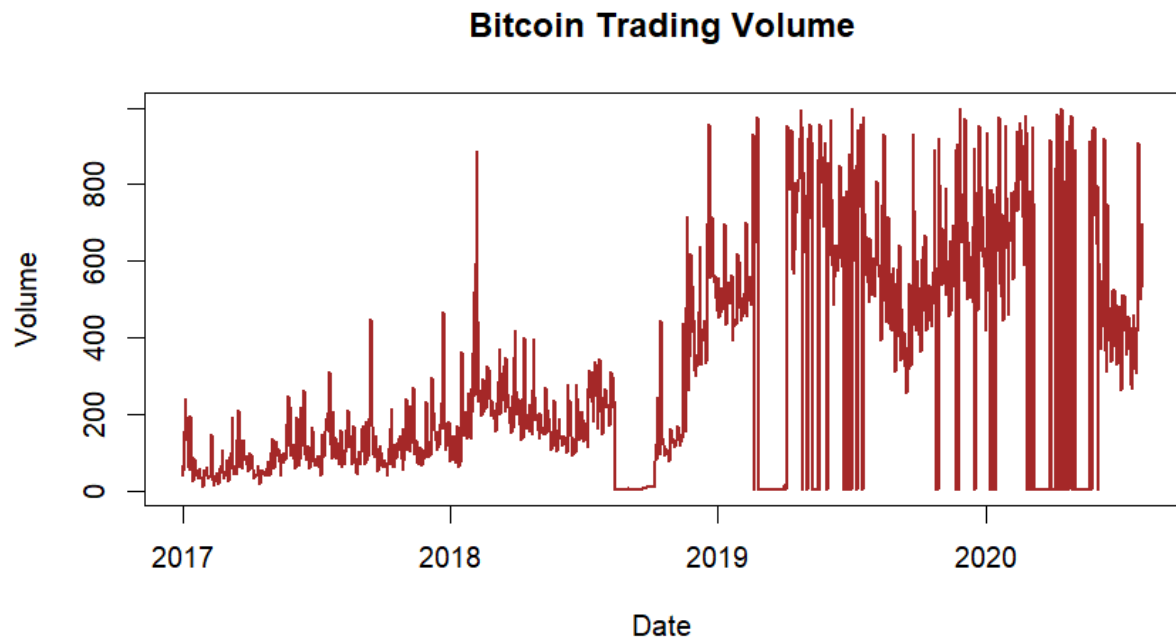
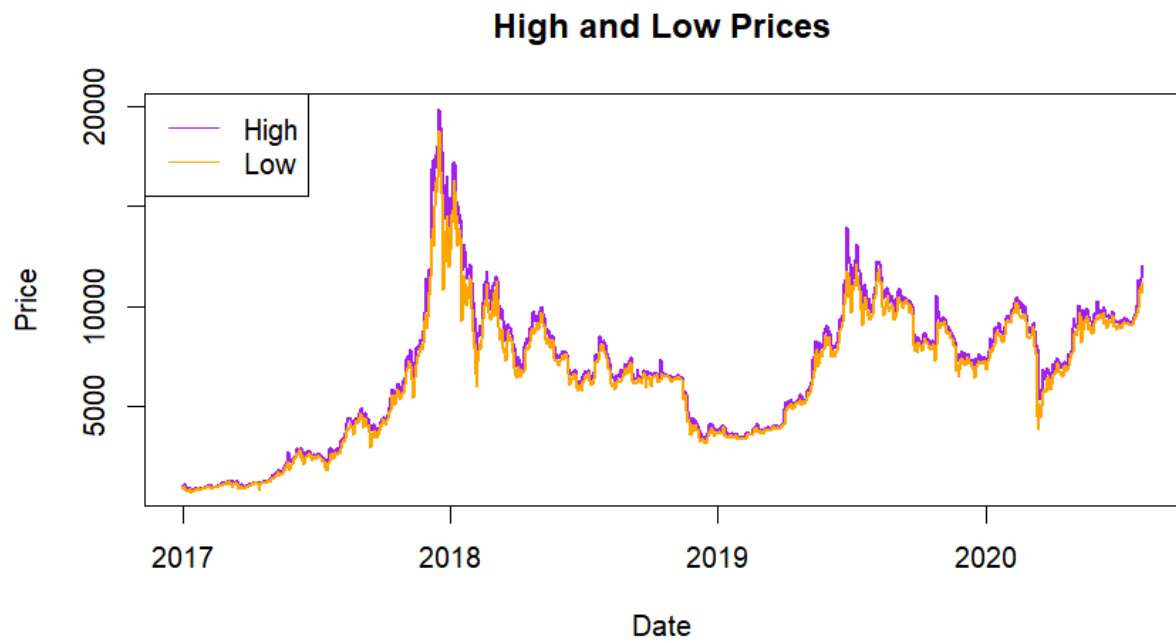
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Graphs:



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References

Github Repo link:

<https://github.com/MainakRepositor/Datasets/blob/master/Bitcoin%20Historical%20Data.csv>

Some more links of good datasets of different topics(github-> free):

<https://github.com/shramos/Awesome-Cybersecurity-Datasets?tab=readme-ov-file#webapps>

<https://github.com/navneet-nmk/Indian-Crime-Data/tree/master>

<https://github.com/prasertcbs/basic-dataset>

<https://github.com/Geo-y20/Uber-Rides-Data-Analysis/tree/main>